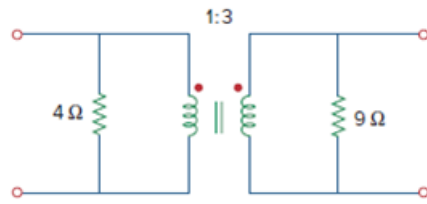


Problem

Determine the z parameters of the two-port in Fig. 19.110.

Figure 19.110



Step-by-step solution

Step 1 of 4

Consider the following two-port network.

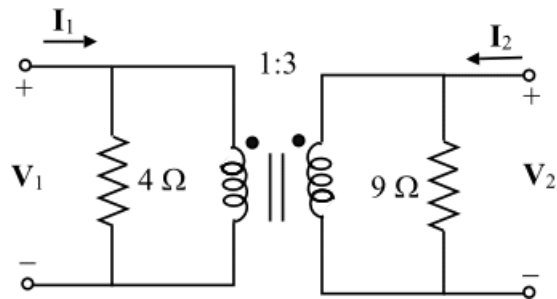


Figure 1

Step 2 of 4

The transformation ratio, $n = 3$

Define z parameters as follows.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

$$\mathbf{z}_{11} = \left. \frac{\mathbf{V}_1}{\mathbf{I}_1} \right|_{\mathbf{I}_2=0}$$

$$\mathbf{z}_{12} = \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{I}_1=0}$$

$$\mathbf{z}_{21} = \left. \frac{\mathbf{V}_2}{\mathbf{I}_1} \right|_{\mathbf{I}_2=0}$$

$$\mathbf{z}_{22} = \left. \frac{\mathbf{V}_2}{\mathbf{I}_2} \right|_{\mathbf{I}_1=0}$$

To determine $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, apply voltage source $\mathbf{V}_1$ and leave the output port open.

Determine the secondary resistance referred to the primary side.

$$\begin{aligned} R_{21} &= \frac{R_2}{n^2} \\ &= \frac{9}{3^2} \\ &= 1 \, \Omega \end{aligned}$$

Draw the transformer circuit referred to the primary side with secondary open.

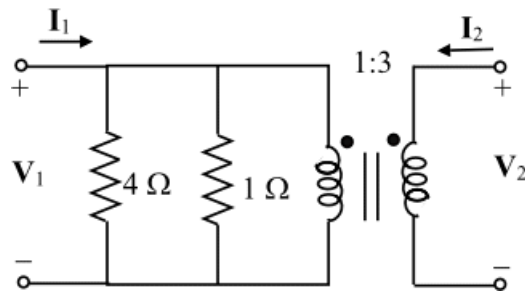


Figure 2

Apply Kirchhoff's voltage law to the primary loop.

$$\begin{aligned}V_1 &= (4 \parallel 1) \mathbf{I}_1 \\&= \frac{(4)(1)}{4+1} \mathbf{I}_1 \\&= \frac{4}{5} \mathbf{I}_1\end{aligned}$$

$$\begin{aligned}\frac{V_1}{\mathbf{I}_1} &= \frac{4}{5} \\&= 0.8\end{aligned}$$

$$\text{Thus, } \mathbf{z}_{11} = \left. \frac{V_1}{\mathbf{I}_1} \right|_{\mathbf{I}_2=0} = 0.8$$

Determine the secondary voltage.

$$\begin{aligned}V_2 &= nV_1 \\&= 3V_1\end{aligned}$$

Substitute $0.8\mathbf{I}_1$ for V_1 .

$$\begin{aligned}V_2 &= (3)(0.8\mathbf{I}_1) \\&= 2.4\mathbf{I}_1\end{aligned}$$

$$\text{Thus, } \mathbf{z}_{21} = \left. \frac{V_2}{\mathbf{I}_1} \right|_{\mathbf{I}_2=0} = 2.4$$

Step 4 of 4

To determine z_{12} and z_{22} , apply voltage source V_2 to the output port and leave the input port open.

Determine the primary resistance referred to the secondary side.

$$\begin{aligned} R_{12} &= n^2 R_1 \\ &= (3^2) 4 \\ &= 36 \Omega \end{aligned}$$

Draw the transformer circuit referred to the secondary side with primary open.

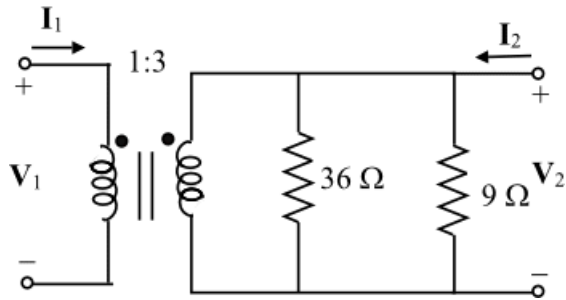


Figure 3

Apply Kirchhoff's voltage law to the secondary loop.

$$\begin{aligned} V_2 &= (36 \parallel 9) I_2 \\ &= \frac{(36)(9)}{36 + 9} I_2 \\ &= \frac{36}{5} I_2 \\ &= 7.2 I_2 \end{aligned}$$

$$\text{Thus, } z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = 7.2$$

Determine the primary voltage.

$$\begin{aligned} V_1 &= \frac{V_2}{n} \\ &= \frac{V_2}{3} \end{aligned}$$

Substitute $7.2 I_2$ for V_2 .

$$\begin{aligned} V_1 &= \frac{7.2 I_2}{3} \\ &= 2.4 I_2 \end{aligned}$$

$$\text{Thus, } z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} = 2.4$$

Thus, the z parameters of the two-port network are $\begin{bmatrix} 0.8 & 2.4 \\ 2.4 & 7.2 \end{bmatrix} \Omega$.